

Are Solar PV Farms Polluting?

Generating solar power using photovoltaic (PV) cells is not a zero-emission technology. However, following all international standards and protocols is considered adequate to curb pollution



RICHA CHAKRAVARTY

Solar energy has long been touted better for the environment than fossil fuels. Considered as one of the 'green' sources of energy, it might become mainstream within the next decade. However, there are fears that solar cells production might release more hazardous gases than fossil fuels. Confirming these fears, Chinese authorities recently suspended production at a solar panel factory after protests by residents who blamed the plant for causing air and water pollution.

In India, on the other hand, the Ministry of Environment and Forests (MoEF), which issues environmental clearances to all projects across the country, has exempted solar photo-

voltic (PV) power projects from the ambit of environmental clearances. The reasoning goes, "Isn't it 'cleaner' to harness the power of the sun?" The obvious answer is 'yes', but only when the right manufacturing processes are followed.

Reasons for pollution

Manufacturing. Solar panel manufacturing generates a number of effluent gases contaminated with silane, trichlorosilane, dichlorosilane and hydrochloric acid (HCl), apart from chemicals like silicon, cadmium compounds, germane and polyvinyl fluoride—all of which are hazardous to the environment and people who are exposed to them. This manufacturing process also requires raw materials that have to be mined—quartz sand for

silicon cells and metal ore for thin-film cells.

These materials are treated at different stages (in the case of silicon cells, these include purification, crystallisation and wafering). Finally, these upgraded materials are manufactured into solar cells and assembled into modules.

All these processes lead to air pollution caused by emission of heavy metals and are highly energy consuming, which brings about more air pollution with emission of heavy metals and release of greenhouse gases. Given the scale at which solar manufacturing is anticipated in India, there is going to be an enormous release of hazardous chemicals and gaseous substances into the environment.

Industry's response. Industry experts and researchers have noted that solar cells produce far fewer air pollutants than conventional fossil-fuel-burning power plants. Making PV cells requires potentially toxic heavy metals such as lead, mercury and cadmium. It even produces greenhouse gases, such as carbon dioxide, which contribute to global warming. Still, researchers found that if people switched from conventional fossil-fuel-burning power plants to solar cells, air pollution would cut down by roughly 90 per cent.

According to Raghunandan, vice president-engineering, Kotak Urja, "Solar PV cells record emission statistics of 32 tonnes per gigawatt hour (GWh) of electricity generated, which is the lowest of all the conventional and non-conventional energy sources. With coal sources emitting a whopping 960 tonnes per GWh, solar PV cells